

CLINICAL REPORT

Modified painless photodynamic therapy for facial multiple actinic keratosis in China: A prospective split-face control study

Jiandan Li MM | Yunfeng Zhang MD | Guolong Zhang MD, PhD |
Linglin Zhang MD | Zhongxia Zhou | Peiru Wang MD, PhD  | Xiuli Wang MD, PhD 

School of Medicine, Institute of
Photomedicine, Shanghai Skin Disease
Hospital, Tongji University, Shanghai,
PR China

Correspondence

Peiru Wang, MD, PhD and Xiuli Wang, MD,
PhD, School of Medicine, Institute of
Photomedicine, Shanghai Skin Disease
Hospital, Tongji University, 200092 Shanghai,
PR China.
Email: wangpeiru@tongji.edu.cn and
wangxiuli_1400023@tongji.edu.cn

Abstract

Background: Aminolevulinic acid photodynamic therapy (ALA-PDT) is an effective treatment for multiple actinic keratosis (AK). However, PDT-induced pain often discontinues the therapy to reduce its efficacy, limiting its application. If modified painless PDT schedule with shorter photosensitizer dressing and higher dose illumination could achieve good efficacy in AK, it is still unknown.

Objectives: To explore the efficacy and pain tolerance of the modified painless PDT (M-PDT) in facial multiple AK.

Methods: A split-face controlled clinical study including 14 patients with facial multiple AK was conducted. The patients received conventional PDT (C-PDT) on the left and M-PDT in the contralateral area. The left area (C-PDT) was illuminated by a red light-emitting diode light (144 J/cm²) after applying the 10% ALA cream for 3 h; the other had illumination for a total light dose of 288 J/cm² after applying the 10% ALA cream for 0.5 h. The primary endpoint was the lesion clearance rate at 1-month postthree sessions of PDT. Secondary endpoints included pain scores, the incidence of adverse events during treatment, and cosmetic outcomes.

Results: At 1 month following three treatments, the total lesion clearance rate was comparable between M-PDT and C-PDT (91.6% vs. 89.0%). While the lesion clearance rate of M-PDT was higher than that of C-PDT in the Grade III lesions (86.5% vs. 72.0%, respectively) ($p < 0.05$). M-PDT achieved a 100% lesion clearance rate for Grade I lesions earlier than C-PDT, with M-PDT treated twice and C-PDT treated thrice. Moreover, the pain score during illumination was significantly lower for M-PDT than for C-PDT ($p < 0.01$). Regarding photoaging, the Global Subjective Skin Aging Assessment score showed that the total and atrophy scores of C-PDT and M-PDT were significantly improved, and M-PDT also reduced discoloration. There was no significant difference in adverse reactions between C-PDT and M-PDT.

Conclusions: M-PDT is comparable to C-PDT's efficacy for treating facial multiple AK, resulting in much lower pain scores.

KEYWORDS

actinic keratosis, pain, photodynamic therapy

INTRODUCTION

Actinic keratosis (AK) is a chronic and progressive precancerous disease, of which 0.1%–16% can progress to cutaneous squamous cell carcinoma (cSCC).^{1,2} The risk of

deterioration increases with the number of skin lesions and the prolonged course of the disease.^{3,4} The malignant rate of multiple AK is much higher than that of single AK, and the malignant rate increases with age.⁵ It is recommended that all patients with AK need treatment and follow-up.

Peiru Wang and Xiuli Wang are co-corresponding authors.

5-Aminolevulinic acid photodynamic therapy (ALA-PDT) is one of the first-line therapy for AK, especially for the head and face, multiple lesions, or AK on large areas.⁶ It somewhat prevents the recurrence of AK and other photoaging skin diseases.^{7,8} The European guideline also recommends PDT treating single or multiple Grade I–II AK (Olsen classification) and field cancerization.^{9,10} A network meta-analysis found that ALA-PDT had significant long-term efficacy in AK treatment.¹¹ However, pain is the primary adverse reaction during conventional PDT (C-PDT), which affects the patient's treatment experience and may not be tolerated, especially in multiple AK and facial areas, resulting in lower efficacy. The efficacy of ALA-PDT is generally thought to be related to reactive oxygen species (ROS) production by photosensitizer during the illumination of PDT.¹² Causes of pain during PDT are also associated with free radicals produced by protoporphyrin IX (PpIX) fluorescence and fluence rate.^{13,14} So, is there a direct relationship between pain and efficacy? Whether the pain can be reduced only at the expense of effectiveness.

To alleviate pain, many clinical trials have explored reducing light energy, lower photosensitizer concentrations, or both.^{14–17} Our previous studies attempted to achieve painlessness by using modified PDT (M-PDT) by reducing the incubation time of ALA and increasing the light dose. Surprisingly, M-PDT has shown excellent efficacy and reduced pain in treating moderate to severe acne vulgaris and condyloma acuminatum.^{18,19} What is more, M-PDT with accumulative higher dose illumination was even more effective than C-PDT for larger cSCC tumors in animal studies.²⁰

Based on these doubts, to better understand M-PDT application value in facial multiple AK, our research aims to conduct a clinical study to compare the efficacy and pain tolerance of M-PDT and C-PDT for AK.

MATERIALS AND METHODS

Research design

This research was designed as a prospective, split-face controlled, subject, and observer-blinded clinical study performed from May 2020 to May 2021 in the Department of Photomedicine, Shanghai Dermatology Hospital. The study was approved by the Ethics Committee of Shanghai Dermatology Hospital (2020-09(ke)) and registered on the Chinese Clinical Trial Registry (no. ChiCTR2000033064). Written informed consent and photo consent were obtained from all patients.

Subjects

Subjects of either sex, aged over 18 years, with the clinical diagnosis of AK Olsen grade I–III on the face

and a lesion number of more than six, were included in the study. Exclusion criteria were a history of photosensitive diseases, internal severe diseases, psychological and mental diseases, infectious diseases or pregnant and lactating women, and any known intolerance to ALA. Patients with clinically and/or pathologically proven tumor invasion to other organs or tissues were excluded. Patients were also excluded if they had received ALA-PDT, topical retinoids or 5-fluorouracil, cryotherapy, or imiquimod at least 4 weeks before the study began for AKs. During the study, using mild skin cleansers and sunscreens was permitted.

Treatment protocol

VISIA system (including 6500-K standard white light and polarized light source) was used to record images. Ten percent fresh ALA cream was prepared using ALA powder (Shanghai Fudan-Zhangjiang Bio-Pharmaceutical Co., Ltd.). The light-emitting diode (LED) red light ($630 \pm \text{nm}$) was used for irradiation and was adjusted to deliver a power density of 40 mW/cm^2 (Kernel Medical Equipment Co., Ltd.). The specific operation methods used were as follows. The scales and crusts were pretreated with plum-blossom needles (Chinese dermal needles, seven stars needles) to enhance ALA transdermal penetration on both sides. This hand tool comprises seven needles, and needling was done by tapping the skin vertically from 5 cm high above quickly 3–5 times after regular disinfection. C-PDT side: 10% ALA cream was incubated on the left facial lesions for 3 h and wiped off before irradiation (144 J/cm^2 , 60 min). M-PDT side: 10% ALA cream was incubated on the right facial lesions for 0.5 h and irradiation (288 J/cm^2 , 120 min). After the C-PDT treatment completed, the M-PDT side was still externally applied with ALA cream. And the unexposed side was shielded with a black cloth when the contralateral side was irradiated with red light. After treatment, the patients were instructed to avoid strong sunlight for at least 2 days. Each patient received three treatments, with each treatment spaced 2 weeks apart. A final lesion count was performed 1 month after the last treatment. After the trial ended, if the skin lesions were not cured, PDT or another therapy was performed until the skin lesions cleared.

Clinical evaluation

AK lesions of each face were recorded using the VISIA image system and standard digital photograph system at each treatment session and follow-up visit. Two blinded dermatologists were involved in clinical efficacy and image data analysis. AK of each face was marked in charts to document their exact location. The number and grade of lesions were counted before each treatment. Lesion clearance was defined as lesions no longer visible and palpable. The lesion clearance rate = $[\text{AK}_o - \text{AK}_p] /$

$AK_o \times 100\%$, where AK_o is the number of lesions at baseline, and AK_p is the number of lesions posttherapy.

Photoaging evaluation

Two dermatologists who did not participate in the clinical trial followed the criteria Global Subjective Skin Aging Assessment (GS2A2) score was performed on the normalized photos and VISIA skin images. GS2A2 includes atrophy (deep/superficial wrinkles, eye bags, lax appearance, etc.), discoloration (freckles, lentigines, melasma, etc.), and malignant skin lesions.²¹

Pain tolerance and safety evaluation

The maximal pain was evaluated at the first PDT treatment. Pain intensity was assessed using the visual analog scale (VAS), with a score ranging from 0–3 points (mild pain), 4–6 points (moderate pain), and 7–10 points (severe pain). The adverse events such as erythema, pruritus, scab, skin burning sensation, and duration of objective symptoms were evaluated and recorded in detail during the whole treatment and follow-up.

Statistical analysis

Statistical analysis was conducted using the Statistical Package for the Social Sciences version 25.0 software (IBM Corporation). Baseline characteristics were compared by the Wilcoxon signed rank test. A χ^2 was performed for the lesion clearance rate after treatment. The linear trend χ^2 test was used to analyze the trend of the lesion clearance rate with the number of treatments. The independent-sample *t*-test or Mann–Whitney *U* test was used to compare the pain score and GS2A2 score between groups. $p < 0.05$ indicated that the difference was statistically significant.

RESULTS

Subject demographics

A total of 14 subjects were screened, including two males and 12 females (Figure 1). The mean age was 75.75 ± 10.10 years (range: 50–91 years). As shown in Table 1, the subjects have a long duration of AK (mean 67 ± 48.30 months). Both treated sides were symmetrical and had similar lesions (202 and 155 for the M-PDT and C-PDT sides, respectively). Regarding AK severity, 18.8% of Grade I lesion and 44.6% of Grade II lesion were treated with C-PDT compared with 16.1% of Grade I lesion and 51.6% of Grade II lesion treated with M-PDT.

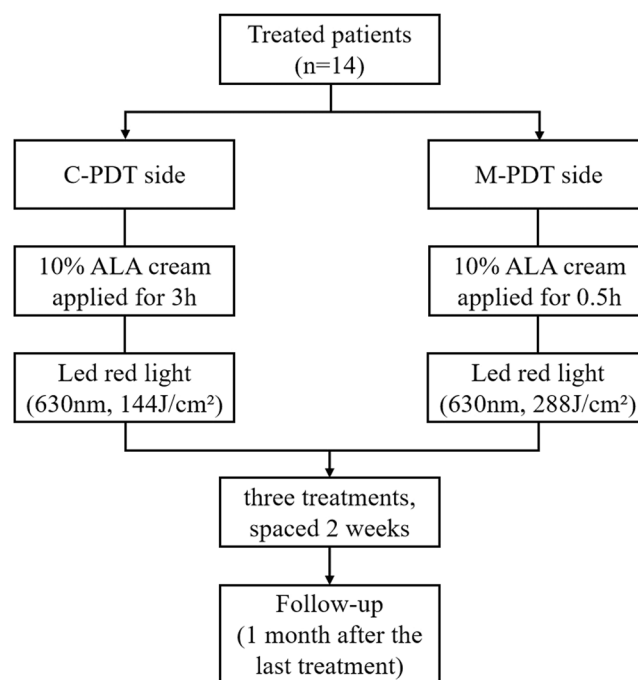


FIGURE 1 Flow chart. ALA, 5-aminolevulinic acid; C-PDT, conventional photodynamic therapy; M-PDT, modified painless photodynamic therapy.

TABLE 1 Patient demographics.

Patient demographics	
Gender (<i>n</i> = 14)	
Male	2
Female	12
Age (years)	
Mean $\pm$ SD	75.75 ± 10.10
Min–max	50–91
Fitzpatrick skin phototype	
II	1
III	10
IV	3
Duration of actinic keratosis (months)	
Mean $\pm$ SD	67 ± 48.30
Min–max	3–144

Abbreviations: max, maximum; min, minimum.

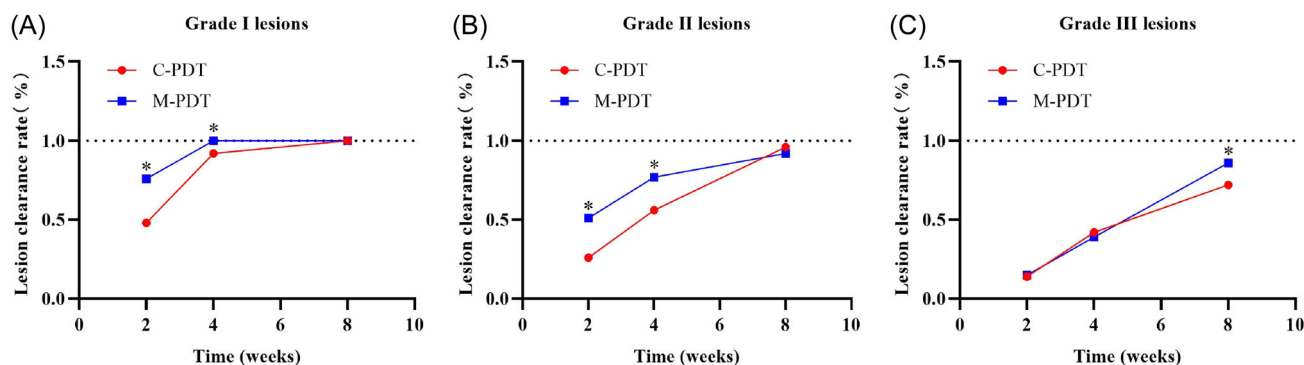
Efficacy

Lesion grades and response rates are presented in Table 2 and Figure 2. Two weeks after the first and the second PDT treatment, the lesion clearance rate of the total between the C-PDT group and the M-PDT group was

TABLE 2 Lesion response of total lesions in the C-PDT group and M-PDT group.

	Baseline	2 Weeks after one session (%)	2 Weeks after two sessions (%)	1 Month after three sessions (%)	<i>p</i> -Value ^a
Side of C-PDT	155	40 (25.8)	87 (56.1)	138 (89.0)	<0.001
Side of M-PDT	202	86 (42.6)	136 (67.3)	185 (91.6)	<0.001
<i>p</i> -Value		0.001	0.030	0.416	

Abbreviations: C-PDT, conventional photodynamic therapy; M-PDT, modified painless photodynamic therapy.

^aLinear trend χ^2 test.**FIGURE 2** Comparison of the lesion clearance rates of (A) Grade I, (B) Grade II, and (C) Grade III lesions between the C-PDT group and M-PDT group. C-PDT, conventional photodynamic therapy; M-PDT, modified painless photodynamic therapy. * $p < 0.05$.

statistically significant ($p < 0.05$). One month after three treatment sessions, the lesion clearance rate of M-PDT was comparable to that of C-PDT ($p > 0.05$). There was a linear increase in the response rates and the number of treatments in both. For the Grade I lesions, the lesion clearance rate of the C-PDT group in two treatments was 48% and 84%, and the M-PDT group was 67.9% and 100%, respectively ($p < 0.05$). M-PDT achieved a 100% lesion clearance rate earlier than C-PDT, with M-PDT treated twice and C-PDT treated thrice. For the Grade II lesions, after the first and the second PDT treatment, the M-PDT group's clearance rate differs from the C-PDT group (51.1% vs. 26.6%, 76.7% vs. 56.3%; $p < 0.05$). Moreover, As the Grade III lesion, the lesion clearance rate in the C-PDT and M-PDT groups were 72.0% and 86.5%, respectively, showing significant differences ($p = 0.045$). The representative clinical manifestations are shown in Figure 3.

Improvement in photoaging

There was an improvement in photoaging skin 1 month after the third PDT treatment. The GS2A2 score was reduced to varying degrees, including the total and atrophy scores. M-PDT also reduced discoloration

(Figure 4). However, the two groups had no statistically significant difference in photoaging improvement.

Pain tolerance and safety

Pain tolerance

M-PDT was almost painless during the first PDT session, with all pain scores ranging from 0 to 2. As shown in Figure 5, the dark bar showed the pain during C-PDT. The pain showed up once the light illuminated, reaching VAS 8 points in 3–5 min, then falling back a bit but still higher than VAS 4 points. The pain lasts during the entire illumination. Patients have to interrupt illumination to alleviate pain. The light bar showed the pain during M-PDT. The pain showed up usually after 7 min and kept no more than VAS 2 points, and the patients could tolerant as long as 2 h illumination. The pain during PDT treatment was significantly lower for M-PDT than for C-PDT ($p < 0.01$).

Safety

During treatment, adverse reactions were observed on both sides of the face, including erythema, pruritus, scab,



FIGURE 3 Treatment of facial multiple actinic keratosis: changes in skin lesions at different times after C-PDT and M-PDT treatment. C-PDT, conventional photodynamic therapy; M-PDT, modified photodynamic therapy.

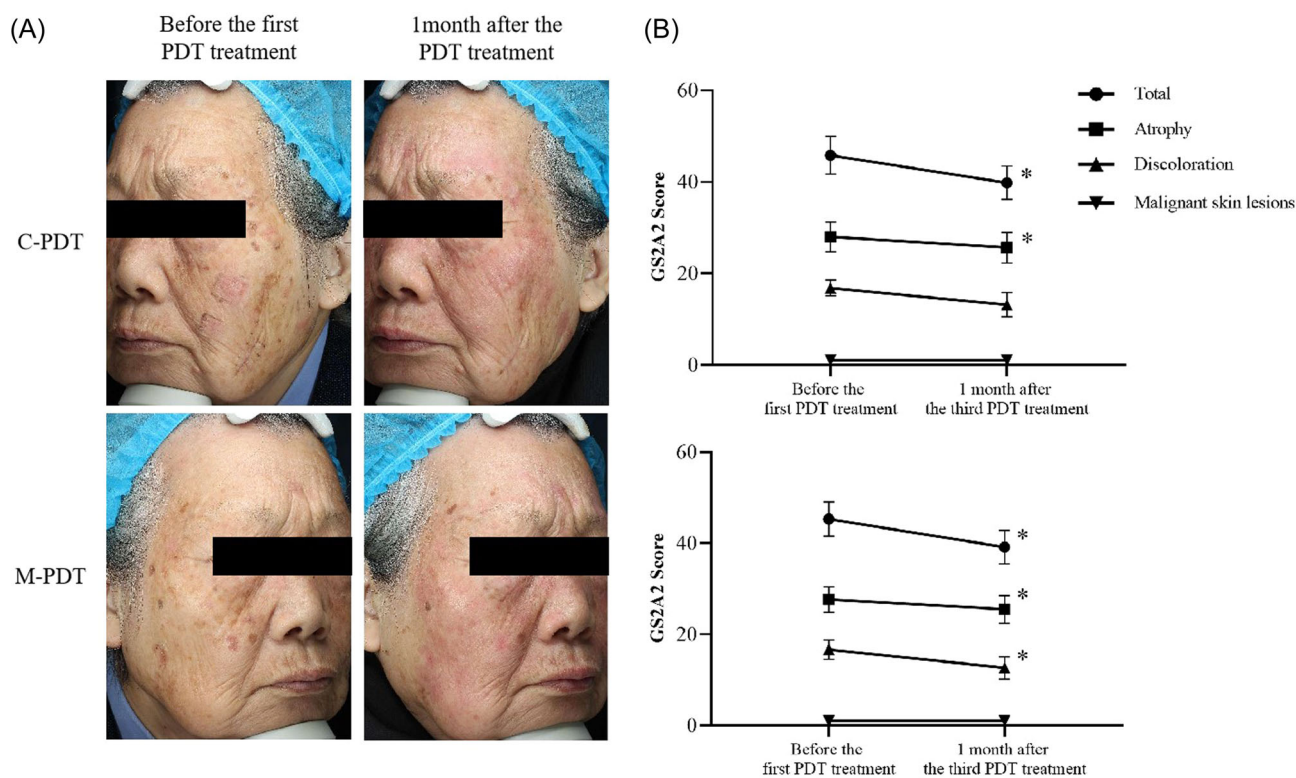


FIGURE 4 The GS2A2 score before and 1 month after the third PDT treatment. C-PDT, conventional photodynamic therapy; GS2A2, Global Subjective Skin Aging Assessment; M-PDT, modified photodynamic therapy. * $p < 0.05$.

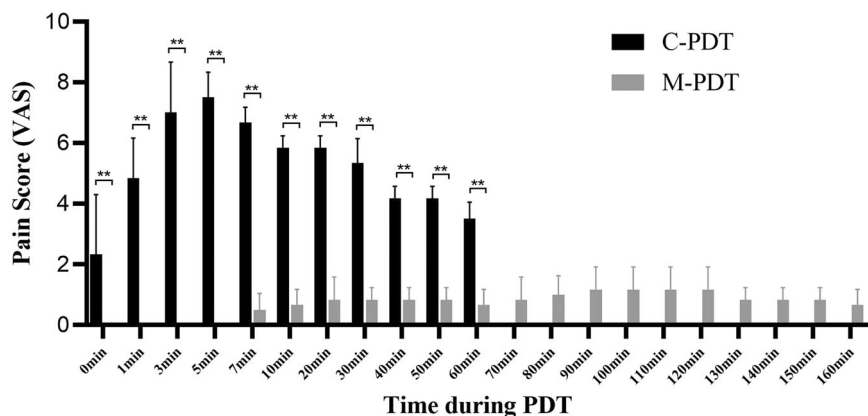


FIGURE 5 The pain score during the first PDT treatment. C-PDT, conventional photodynamic therapy; M-PDT, modified photodynamic therapy; VAS, visual analog scale. $**p < 0.01$.

TABLE 3 Adverse effects.

	Side of C-PDT (<i>n</i> = 14) Subject (<i>n</i> [%], mean duration)	Side of M-PDT (<i>n</i> = 14) Subject (<i>n</i> [%], mean duration)
ALL AEs	14 (100)	14 (100)
Related AEs	14 (100)	14 (100)
Serious AEs	0	0
Erythema	14 (100), 3.8 days	14 (100), 3.8 days
Pruritus	12 (86), 1.6 days	9 (64), 1.0 days
Scab	14 (100), 10.8 days	14 (100), 10.8 days
Skin burning sensation	0	0
Skin edema	2 (14), 1.0 days	2 (14), 1.0 days
Dry	14 (100), 3.3 days	14 (100), 3.3 days
Oozing	2 (14), 1.0 days	2 (14), 1.0 days
Phototoxicity reaction	0	0
Scar	0	0
Hypopigmentation	0	0
Hyperpigmentation	14 (100), 8.3 weeks	14 (100), 8.3 weeks

Abbreviations: AE, adverse event; C-PDT, conventional photodynamic therapy; M-PDT, modified photodynamic therapy.

edema, dryness, and so forth. Erythema, scab, dryness, and pigmentation were observed on both sides of C-PDT and M-PDT, and the duration was the same on both sides. Pruritus occurred in 83% of the C-PDT side, compared with 67% of the M-PDT side, and the C-PDT side lasted longer than the M-PDT side. Two persons occurred for skin edema on both the C-PDT and the M-PDT sides. No patients in either group experienced the skin-burning sensation, phototoxicity reaction, scar, hypopigmentation, or other significant adverse reactions (Table 3).

DISCUSSION

The present study demonstrates that M-PDT may be an effective, virtually painless, and convenient alternative to C-PDT for treating facial multiple AK. The primary endpoint analysis showed that M-PDT was significantly higher than C-PDT in the Grade III lesion, with a 1-month lesion clearance rate was 72.0% in C-PDT and 86.5% in M-PDT. Furthermore, M-PDT showed better pain tolerance and higher patient compliance than C-PDT in pain. M-PDT was preferred in patients, with all mean pain scores less than one on M-PDT and more than four on C-PDT.

Many trials exploring painless PDT have used ALA or MAL-PDT as C-PDT, and the AK lesion clearance rate after one session of PDT treatment was approximately 55.4%–80.7% in both groups.^{22–24} However, the lesion clearance rate after the first PDT session in this study was lower than that in previous studies, which was 25.8% for C-PDT and 42.6% for M-PDT. It may be due to the larger proportion of Grade II/III AK lesions (the total number of Grade II and Grade III more than 80%) in this study and the higher proportion of Grade III lesions that were hypertrophic. Therefore, hypertrophic AK lesions may make treatment more difficult.

In the study, there was no significant difference in the lesion clearance rate between C-PDT and M-PDT in the total lesion of AK 1 month after three sessions PDT. However, the lesion clearance rate of M-PDT on Grade III lesions was more ideal than C-PDT. As well as we observed that the lesion clearance rate of M-PDT reached 100% clearance earlier than C-PDT for Grade I lesions and a higher clearance rate for Grade II lesions. M-PDT was comparable to C-PDT in treating multiple AK on the face, similar to animal models.²⁰ Based on a mouse model of squamous cell carcinoma treated with M-PDT, M-PDT produced more cumulative ROS than C-PDT during treatment. In addition, M-PDT has a more substantial inhibitory effect on the signaling

pathways of tumor proliferation, such as signal transducer and activator of transcription 3, extracellular signal-regulated kinase 1/2, and mechanistic target of rapamycin. Moreover, immune cell populations can be recruited to mediate adaptive immunity to inhibit cSCC progression.²⁰ The therapeutic mechanism of M-PDT may not just be apoptosis and necrosis induced by ROS. Other unconventional cell death pathways, such as pyroptosis, necrotic apoptosis, and ferroptosis, could exist, which need further investigation. In addition, a PDT-induced immune response might play a role.^{25,26}

We noticed that both C-PDT and M-PDT showed excellent skin rejuvenation. According to the GS2A2 score, facial atrophy and color changes were improved at the 1-month follow-up, and there was no significant difference between the two groups. This result is consistent with previous clinical trials on photodynamic rejuvenation. ALA-PDT can improve the density and arrangement of collagen and elastic fibers in photodamaged skin.²⁷

During the treatment, the pain VAS of C-PDT was above four and peaked at 5 min after treatment. The average score of M-PDT was below one, and a VAS score of less than two is considered painless. PDT pain is thought to be mediated by free radicals produced by PpIX; ROS can directly stimulate nerve endings or mediate pain through inflammatory by-products.²⁸ We hypothesized that M-PDT would be similar to daylight PDT in that a small amount of continuous PpIX production and light activation rather than a large amount of rapid activation would reduce pain perception.¹⁴

Numerous studies have explored the intrinsic and extrinsic factors associated with PDT pain. The internal factors include disease type, lesion site, gender, and lesion area. PDT treatment of skin lesions on the head and face is more intense than in other locations, and the larger the lesion area, the more painful.²⁹ In addition, extrinsic factors, including photosensitizer, fluence rate, and light dose, are thought to be more closely related to pain in PDT. By analyzing the literature on the correlation between fluence rate and pain intensity, Wang et al.³⁰ proposed a threshold theory for pain during PDT: it is positively correlated with energy rate and dose below a certain threshold (rate: $\sim 60 \text{ mW/cm}^2$, dose: $\sim 50 \text{ J/cm}^2$), beyond which pain intensity no longer increases. The same therapeutic effect can be achieved by decreasing the fluence rate while extending the irradiation time, thus allowing slow and continuous production of ROS to reduce pain. Based on this theory, Apalla et al.⁸ provided patients with a low-irradiance PDT (li-PDT) protocol, with a total light dose of 37 J/cm^2 and a reduction in irradiance of 25% of the traditional irradiance, and a fourfold increase in illumination time. Results showed that li-PDT patients had

significantly less pain (VAS score 2.8 vs. 7.6) and fewer treatment interruptions; no significant difference in treatment effect was found.¹⁵

Previous studies have demonstrated that pain can be reduced by shortening the drug–light interval (DLI) and prolonging the light time.²⁴ For example, Mordon et al.²² showed that one area was illuminated with a red LED light (37 J/cm^2) after 3 h incubation of the MAL dressing. The other side was irradiated for 2.5 h with the new device half an hour after applying MAL ointment. Both sides' lesion complete response rates at 6-month follow-up were similar (94.2% vs. 94.9%). However, the pain scores for 30-min DLIs were significantly lower than those for 3-h (0.3 and 7.4, $p < 0.001$).²² Maire et al.³¹ performed MAL-PDT in 38 male patients with scalp AK lesions with artificial daylight (white LED) almost immediately (3–5 min) after the application of MAL cream for 2.5 h (light dose: 26.1 J/cm^2). Thirty-three (87%) of 38 had no pain, and the remaining five (13%) had a pain grade of one. Unlike the above studies, we increased the light dose while shortening the drug's light time as M-PDT. The results showed that based on pain relief, the clearance rate of skin lesions also increased to a certain extent. Previous studies by our team found that M-PDT significantly reduced the expression of TRPV1, PAR2, and substance P compared with C-PDT to minimize the occurrence of pain.²⁰ Our previous research suspected that in the C-PDT, the accumulation of ROS per unit of time destroys tumor cells and stimulates nerve endings to produce pain. On the contrary, PpIX production of M-PDT was slow, and the amount of ROS accumulated per unit of time was insufficient to activate nerve endings to produce pain.²⁰ On the other hand, M-PDT does not update equipment that is easy to use clinically. But prolonged bright light requires special eye protection.

Regarding side effects, both C-PDT and M-PDT were safe in our study, with a few mild adverse reactions observed. Erythema, scab, dryness, and hyperpigmentation were observed in all subjects. Some subjects also had pruritus, skin edema, and oozing. The symptoms are usually mild and transient, do not cause severe discomfort, and can be relieved without additional treatment. None of the subjects experienced severe adverse reactions.

Due to this study's limited number of cases, more studies are needed, especially for regional cancer patients.

In conclusion, this study found that the efficacy of M-PDT and C-PDT for AK is comparable, reducing pain during treatment and allowing patients to have a better experience. In other adverse reactions, there was no significant difference between M-PDT and C-PDT. Moreover, both have excellent skin rejuvenation effects.

ACKNOWLEDGMENT

The authors would like to thank the personnel of "Institute of Photomedicine" for their cooperation in this study.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ORCID

Peiru Wang  <https://orcid.org/0000-0003-0006-4269>

Xiuli Wang  <https://orcid.org/0000-0003-0144-7480>

REFERENCES

- de Oliveira ECV, da Motta VRV, Pantoja PC, Ilha CSO, Magalhães RF, Galadari H, et al. Actinic keratosis—review for clinical practice. *Int J Dermatol*. 2019;58(4):400–7.
- Freeman RG. Carcinogenic effects of solar radiation and prevention measures. *Cancer*. 1968;21(6):1114–20.
- Flohil SC, van der Leest RJT, Dowlathshahi EA, Hofman A, de Vries E, Nijsten T. Prevalence of actinic keratosis and its risk factors in the general population: the Rotterdam Study. *J Invest Dermatol*. 2013;133(8):1971–8.
- Ulrich C, Jürgensen JS, Degen A, Hackethal M, Ulrich M, Patel MJ, et al. Prevention of non-melanoma skin cancer in organ transplant patients by regular use of a sunscreen: a 24 months, prospective, case-control study. *Br J Dermatol*. 2009;161:78–84.
- Dodson JM. Malignant potential of actinic keratoses and the controversy over treatment—a patient-oriented perspective. *Arch Dermatol*. 1991;127(7):1029–31.
- Chinese Association of Rehabilitation Dermatology, Photodynamic Therapy Research Center, Chinese Society of Dermatology, Photomedicine Therapeutic Equipment Group, Committee on Skin Disease and Cosmetic Dermatology, China Association of Medical Equipment. Expert consensus on clinical diagnosis and treatment of actinic keratosis in China (2021). *Chin J Dermatol*. 2021;54(12):1048–56.
- Patel G, Armstrong AW, Eisen DB. Efficacy of photodynamic therapy vs other interventions in randomized clinical trials for the treatment of actinic keratoses: a systematic review and meta-analysis. *JAMA Dermatol*. 2014;150(12):1281–8.
- Apalla Z, Sotiriou E, Chovarda E, Lefaki I, Devliotou-Panagiotidou D, Ioannides D. Skin cancer: preventive photodynamic therapy in patients with face and scalp cancerization. A randomized placebo-controlled study. *Br J Dermatol*. 2010;162(1):171–5.
- Heppt MV, Leiter U, Steeb T, Amaral T, Bauer A, Becker JC, et al. S3 guideline for actinic keratosis and cutaneous squamous cell carcinoma—short version, part 1: diagnosis, interventions for actinic keratoses, care structures and quality-of-care indicators. *J Dtsch Dermatol Ges*. 2020;18(3):275–94.
- Werner RN, Stockfleth E, Connolly SM, Correia O, Erdmann R, Foley P, et al. Evidence- and consensus-based (S3) guidelines for the treatment of actinic keratosis—International League of Dermatological Societies in cooperation with the European Dermatology Forum—Short version. *J Eur Acad Dermatol Venereol*. 2015;29(11):2069–79.
- Steeb T, Wessely A, Petzold A, Brinker TJ, Schmitz L, Leiter U, et al. Evaluation of long-term clearance rates of interventions for actinic keratosis: a systematic review and network meta-analysis. *JAMA Dermatol*. 2021;157(9):1066–77.
- Anand S, Ortel BJ, Pereira SP, Hasan T, Maytin EV. Biomodulatory approaches to photodynamic therapy for solid tumors. *Cancer Lett*. 2012;326(1):8–16.
- Wiegell SR, Skiveren J, Philipsen PA, Wulf HC. Pain during photodynamic therapy is associated with protoporphyrin IX fluorescence and fluence rate. *Br J Dermatol*. 2008;158(4):727–33.
- Wiegell SR, Hædersdal M, Philipsen PA, Eriksen P, Enk CD, Wulf HC. Continuous activation of PpIX by daylight is as effective as and less painful than conventional photodynamic therapy for actinic keratoses; a randomized, controlled, single-blinded study. *Br J Dermatol*. 2008;158(4):740–6.
- Gholam P, Bosselmann I, Enk AH, Dick J. Low irradiance compared with conventional photodynamic therapy in the treatment of actinic keratoses. *Photodermatol Photoimmunol Photomed*. 2019;35(2):110–5.
- Apalla Z, Sotiriou E, Panagiotidou D, Lefaki I, Goussi C, Ioannides D. The impact of different fluence rates on pain and clinical outcome in patients with actinic keratoses treated with photodynamic therapy. *Photodermatol Photoimmunol Photomed*. 2011;27(4):181–5.
- Lerche CM, Fabricius S, Philipsen PA, Wulf HC. Correlation between treatment time, photobleaching, inflammation and pain after photodynamic therapy with methyl aminolevulinic acid on tape-stripped skin in healthy volunteers. *Photochem Photobiol Sci*. 2015;14(5):875–82.
- Zhang Y, Zhang H, Zhang L, Wang P, Shi L, Zhang G, et al. Modified 5-aminolevulinic acid photodynamic therapy to reduce pain in the treatment of moderate to severe acne vulgaris: a prospective, randomized, split-face study. *J Am Acad Dermatol*. 2021;84(1):218–20.
- Zhang H, Shi L, Zhang Y, Wang P, Zhang G, Cao Y, et al. Modified photodynamic therapy to minimize pain in the treatment of condylomata acuminata: a prospective, randomized, self-controlled study. *Photodiagn Photodyn Ther*. 2020;32:101915.
- Zeng Q, Zhou C, Zhang Y, Yan G, Wang X. Modified 5-aminolevulinic acid photodynamic therapy reduces pain and improves therapeutic effects in cutaneous squamous cell carcinoma mouse model. *Lasers Surg Med*. 2022;54(5):804–12.
- Buranasirin P, Pongpirul K, Meephansan J. Development of a Global Subjective Skin Aging Assessment score from the perspective of dermatologists. *BMC Res Notes*. 2019;12(1):364.
- Mordon S, Vignion-Dewalle AS, Abi-Rached H, Thecu E, Lecomte F, Vicentini C, et al. The conventional protocol vs. a protocol including illumination with a fabric-based biophotonic device (the Phosistos protocol) in photodynamic therapy for actinic keratosis: a randomized, controlled, noninferiority clinical study. *Brit J Dermatol*. 2020;182(1):76–84.
- Kaw U, Ilyas M, Bullock T, Rittwage L, Riha M, Vidimos A, et al. A regimen to minimize pain during blue light photodynamic therapy of actinic keratoses: bilaterally controlled, randomized trial of simultaneous versus conventional illumination. *J Am Acad Dermatol*. 2020;82(4):862–8.
- Mordon S. Painless and efficient ALA-PDT and MAL-PDT of actinic keratosis can be achieved by drastically reducing the drug-light interval. *Dermatol Ther*. 2020;33(3):e13423.
- Mishchenko T, Balalaeva I, Gorokhova A, Vedunova M, Krysko DV. Which cell death modality wins the contest for photodynamic therapy of cancer? *Cell Death Dis*. 2022;13(5):455.
- Anand S, Yasinchak A, Govande M, Shakya S, Maytin EV. Painless versus conventional photodynamic therapy for treatment of actinic keratosis: comparison of cell death and immune response in a murine model. *Proc SPIE*. 2019;139(5):10860.
- Ji J, Zhang LL, Ding HL, Wang HW, Huang Z, Wang XX, et al. Comparison of 5-aminolevulinic acid photodynamic therapy and red light for treatment of photoaging. *Photodiagn Photodyn Ther*. 2014;11(2):118–21.
- Hackel D, Pflücke D, Neumann A, Viebahn J, Mousa S, Wischmeyer E, et al. The connection of monocytes and reactive oxygen species in pain. *PLoS One*. 2013;8(5):63564.
- Zheng Z, Zhang LL, Shi L, Zhang YF, Wang B, Wu QQ, et al. What is the most relevant factor causing pain during ALA-PDT?

- A multi-center, open clinical pain score research trial of actinic keratosis, acne and condylomata acuminata. *Photodyn Ther*. 2019;26:73–8.
30. Wang B, Shi L, Zhang YF, Zhou Q, Zheng J, Szeimies RM, et al. Gain with no pain? Pain management in dermatological photodynamic therapy. *Br J Dermatol*. 2017;177(3):656–65.
 31. Maire C, Vignion-Dewalle AS, Cartier H, Mordon S. Artificial white light photodynamic therapy for actinic keratosis: a study of 38 patients in private office practice. *J Eur Acad Dermatol Venereol*. 2020;34(4):e165–7.

How to cite this article: Li J, Zhang Y, Zhang G, Zhang L, Zhou Z, Wang P, et al. Modified painless photodynamic therapy for facial multiple actinic keratosis in China: A prospective split-face control study. *Lasers Surg Med*. 2023;55:871–879. <https://doi.org/10.1002/lsm.23728>